



# EN02-OP

## Open-Source Robot End-Effector

# Assembly Guide



Project Repo: [github.com/Westwood-Robotics/EN02-OP](https://github.com/Westwood-Robotics/EN02-OP)  
Website: [www.WestwoodRobotics.io](http://www.WestwoodRobotics.io)

# EN02 BOM Preparation:

## Screws:

- S1: Flat Head TRS-M2.2x12, 28pc
- S2: Flat Head TRS-M2x8, 4pc
- S3: Flat Head TRS-M4x12, 2pc
- S4: Rounded/Button Head TRS-M2.5x6 Screw, 1pc
- S5: Flat Head TRS-M2.5x8, 1pc
- S6: Flat Head TRS-M3x12, 1pc
- S7: Flat Head TRS-M3x8, 7pc
- S8: Flat Head TRS-M2x5, 24pc
- S9: Flat Head TRS-M3x10, 8pc
- S10: Rounded/Socket Head TRS-M4x10, 6pc
- S11: Flat Head TRS-M2.5x24, 2pc

## Dowel Pins:

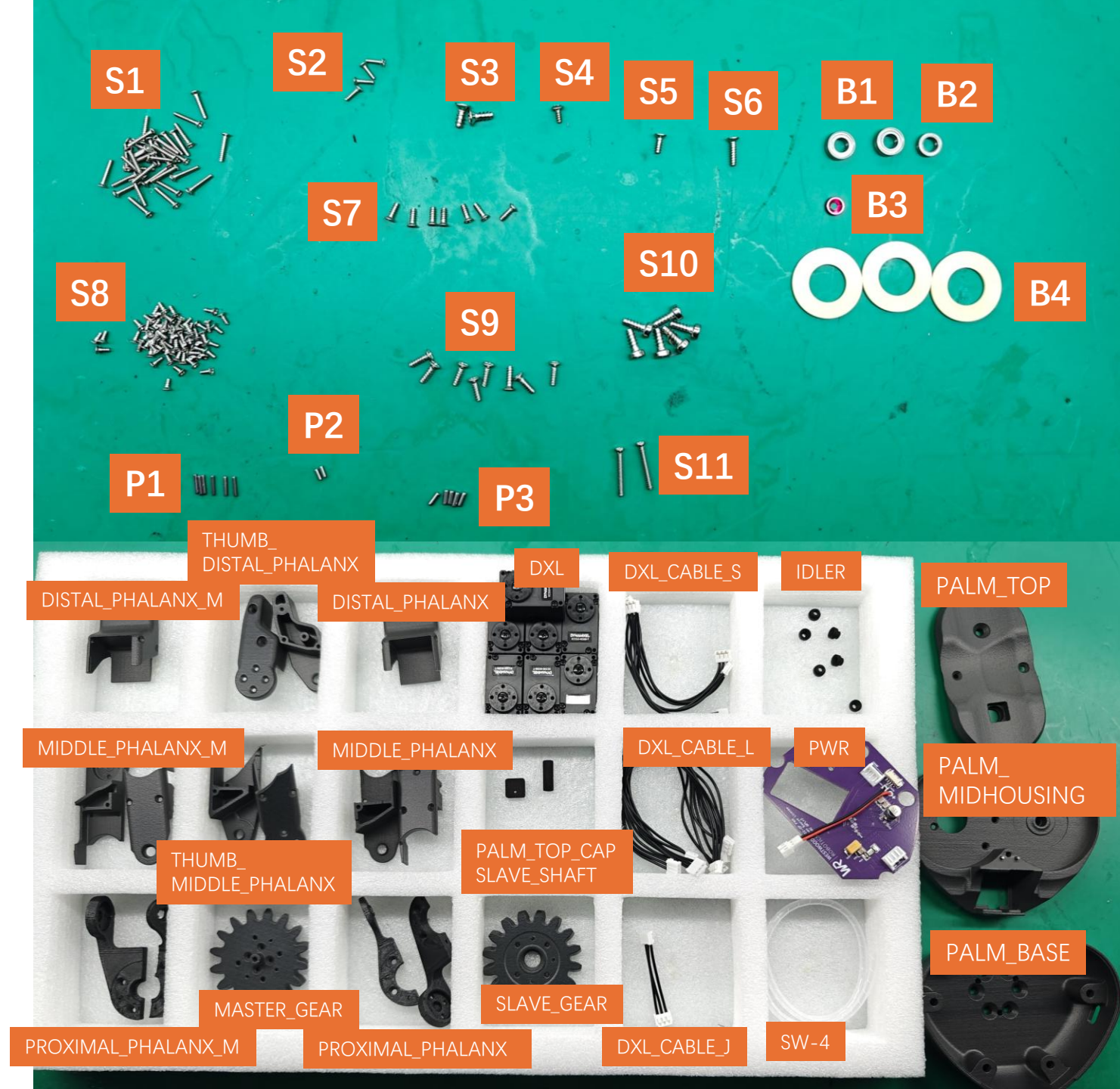
- P1: 1.5x10 Dowel Pin, 6pc
- P2: 1.5x5 Dowel Pin, 2pc
- P3: 1.5x5 Dowel Pin, 5pc

## Bearings & Hardware:

- B1: Bearing 126ZZ, 2pc
- B2: Bearing 106ZZ, 1pc
- B3: Metal Washer For Flat Head Screw, M3, 1pc
- B4: IGUS JTM-1630-015 Thrust Bearing, 3pc

## EN02 Parts:

- Prepare according to **BOM** (Found in project folder).
- PROXIMAL\_PHALANX contains two parts each and must be printed with CF reinforced material.
- Modify the design of PALM\_BASE to fit your mounting interface before printing.
- Solder your preferred power header to PWR board.





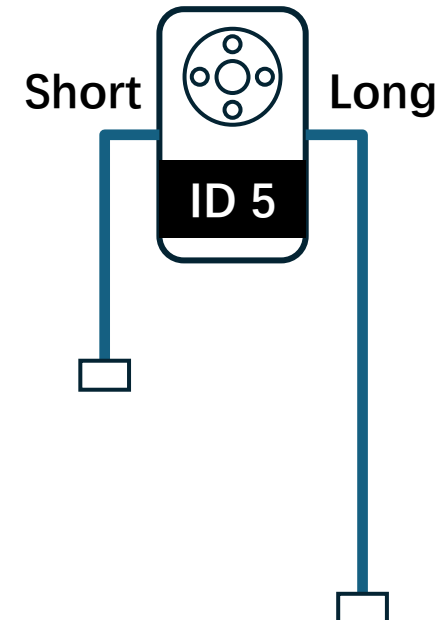
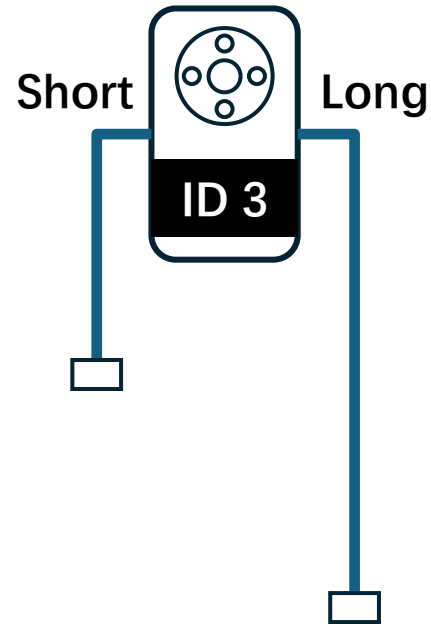
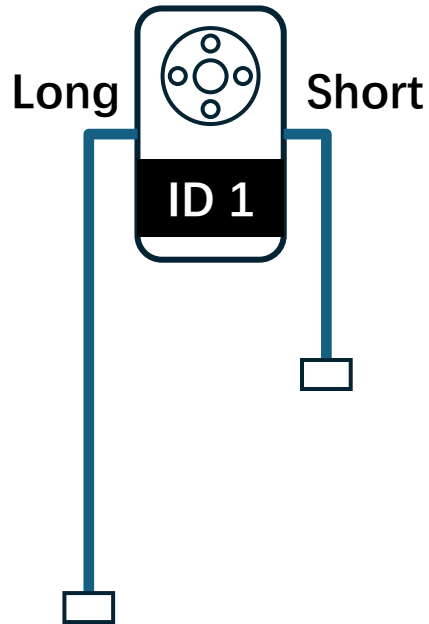
## Step 1.1 Idler Preparation:

Insert one **S7: Flat Head TRS-M3x8** screw into each of the 6 **IDLER**s to create 6 of **IDLER\_w/\_SCREW**



## Step 1.2 DXL Preparation:

1. Set the IDs of the 7 **Dynamixel XC/XL 330** actuators to 1~7. Note that if you are using six **XC-330** and one **XL-330** instead of all 7 identical actuators, make sure to set the **XL-330** actuator's ID to 7.
2. Assemble one **DXL\_CABLE\_S** cable and one **DXL\_CABLE\_L** cable to the **XC/XL 330** actuators with ID 1, 3 and 5, on the corresponding ports exactly as shown below.

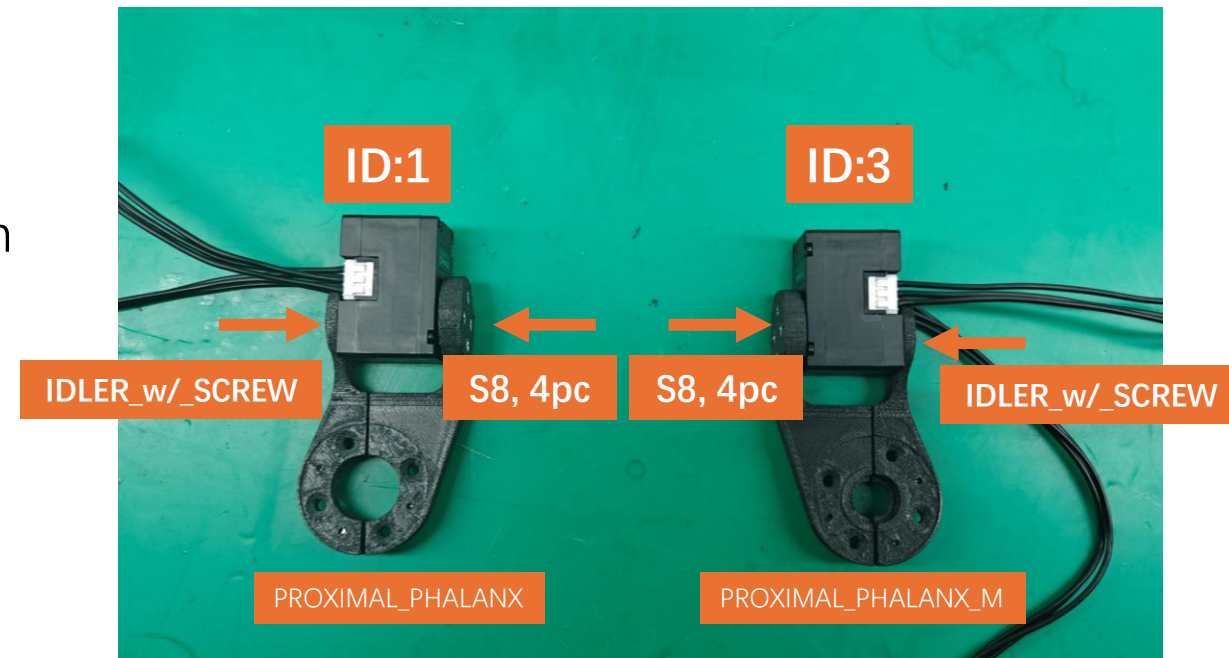
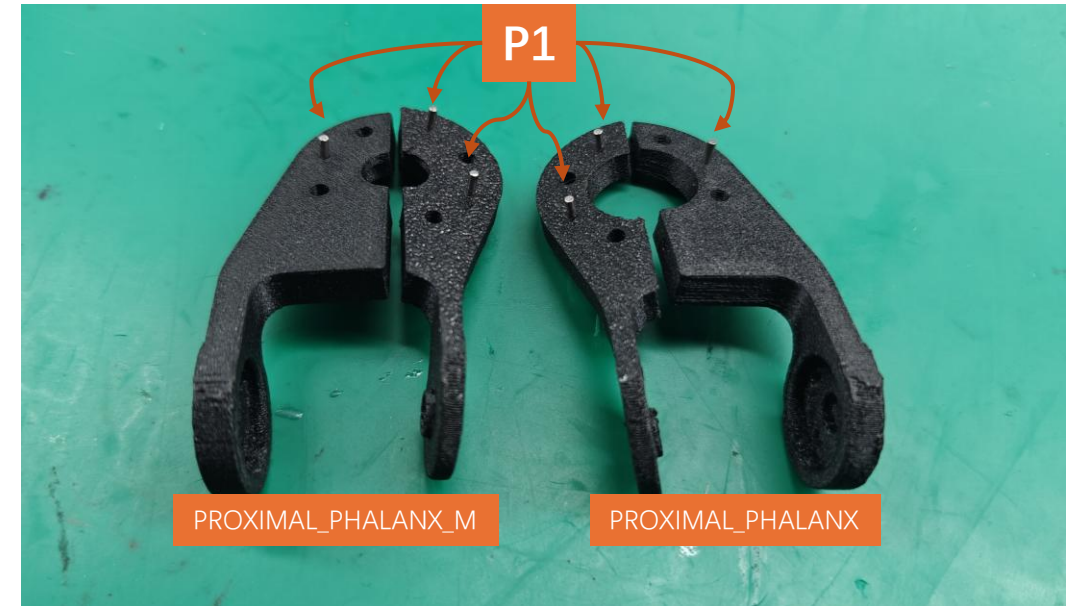


## Step 2 Index Fingers-1:

1. Insert 3 **P1: 1.5x10 Dowel Pin** respectively into the pin holes on **PROXIMAL\_PHALANX\_M** and **PROXIMAL\_PHALANX**
2. Assemble **PROXIMAL\_PHALANX** with **DXL ID1** and **PROXIMAL\_PHALANX\_M** with **DXL ID3** respectively, each using one **IDLER\_w/\_SCREW** and four **S8: Flat Head TRS-M2x5**.

### IMPORTANT:

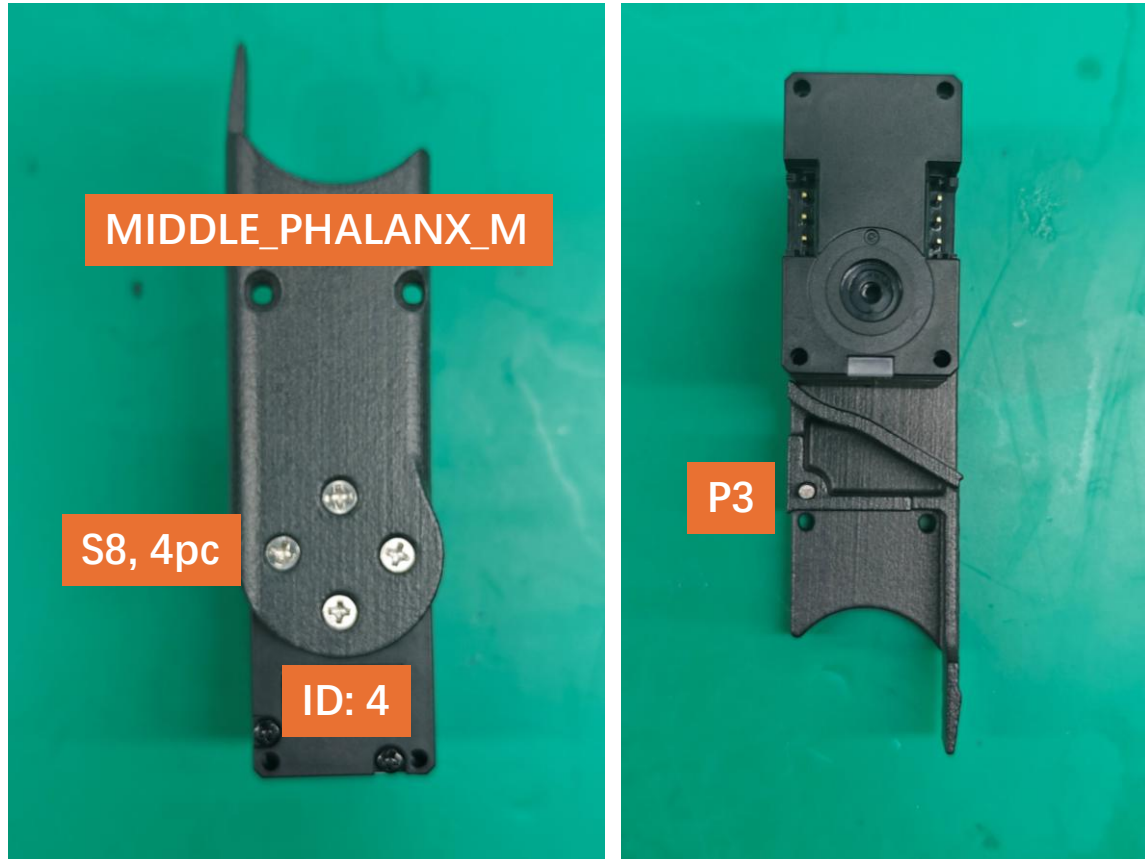
Parts must be assembled in the position as show in the picture on the right, and avoid moving the output shaft of DXL out of its zero position during assembly.





## Step 2 Index Fingers-2:

Assembly half of **MIDDLE\_PHALANX\_M** onto **DXL ID4** using four **S8: Flat Head TRS-M2x5**, and insert one **P3: 1.5x5 Dowel Pin**

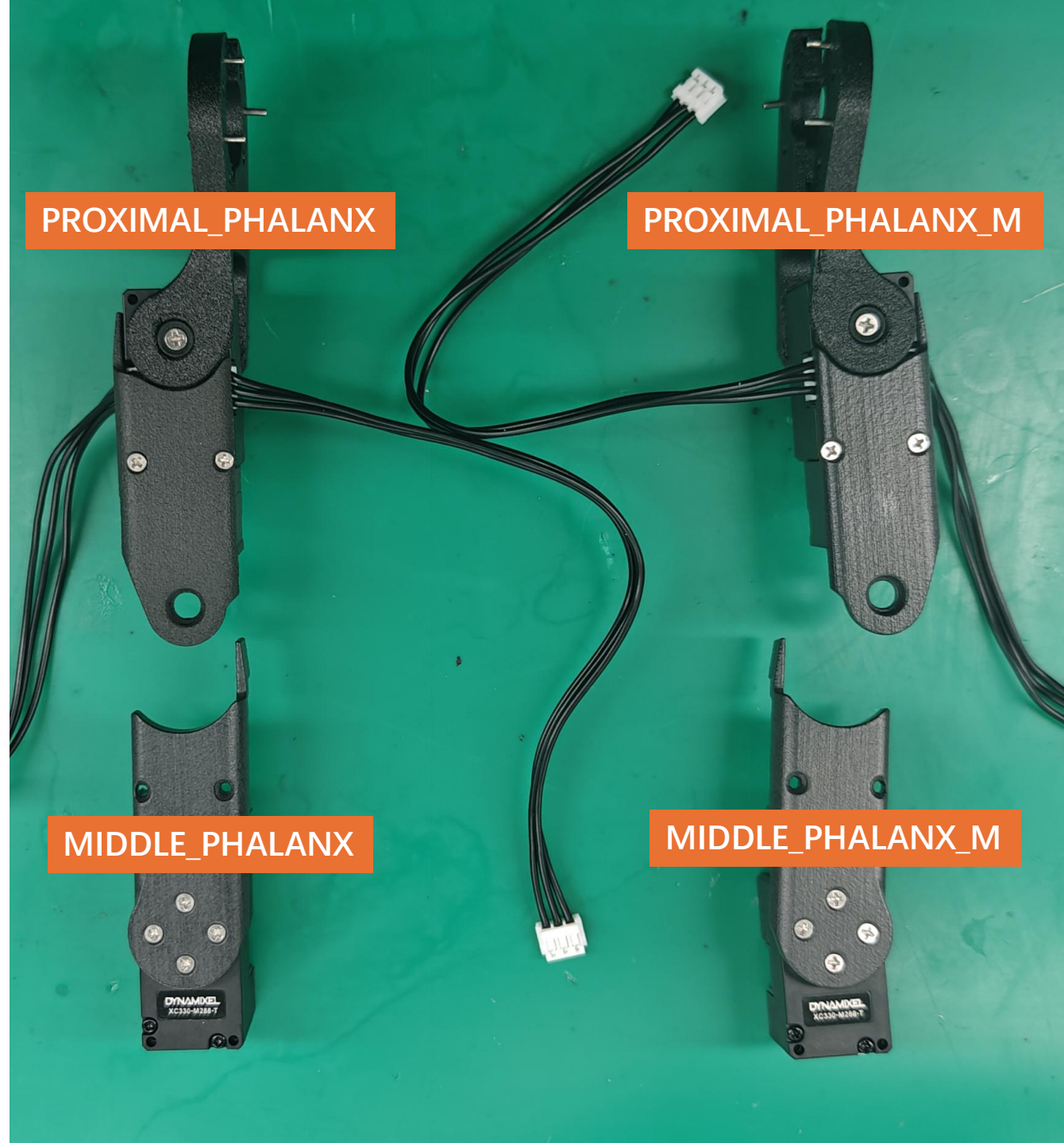


Assembly the other half of **MIDDLE\_PHALANX\_M** onto **DXL ID3** using two **S1: Flat Head TRS-M2.2x12**, as shown in the picture below:



## Step 2 Index Fingers-3:

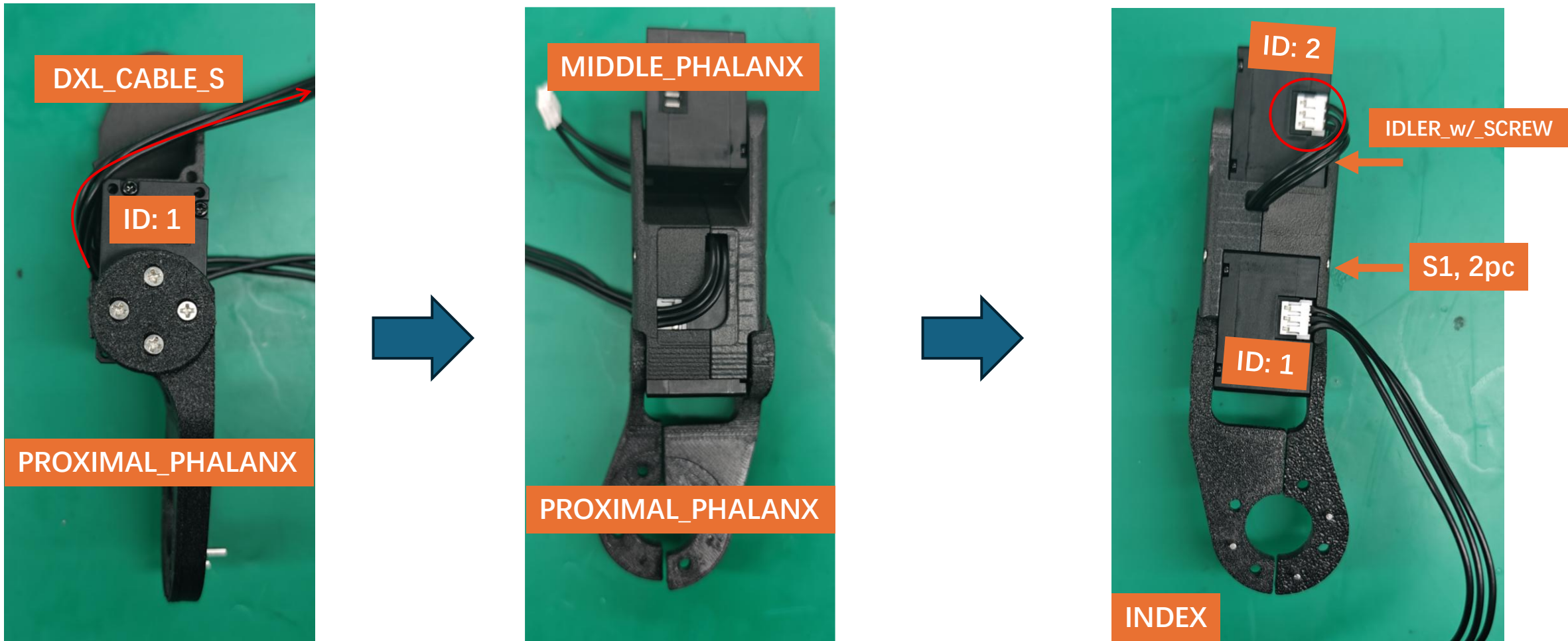
Repeat previous step and assemble **MIDDLE\_PHALANX**, **DXL ID2**, and **DXL ID1**, to get the **MIDDLE\_PHALANX** and **PROXIMAL\_PHALANX** subassemblies.



## Step 2 Index Fingers-4:

Route the **DXL\_CABLE\_S** connected to **DXL ID1** as shown in the picture below, then assemble the **MIDDLE\_PHALANX** subassembly onto the **PROXIMAL\_PHALANX** subassembly and secure with one **IDLER\_w/\_SCREW** and two **S1: Flat Head TRS-M2.2x12** to form the **INDEX** subassembly.

Insert the free header of the **DXL\_CABLE\_S** cable into the nearby port on **DXL ID2**.

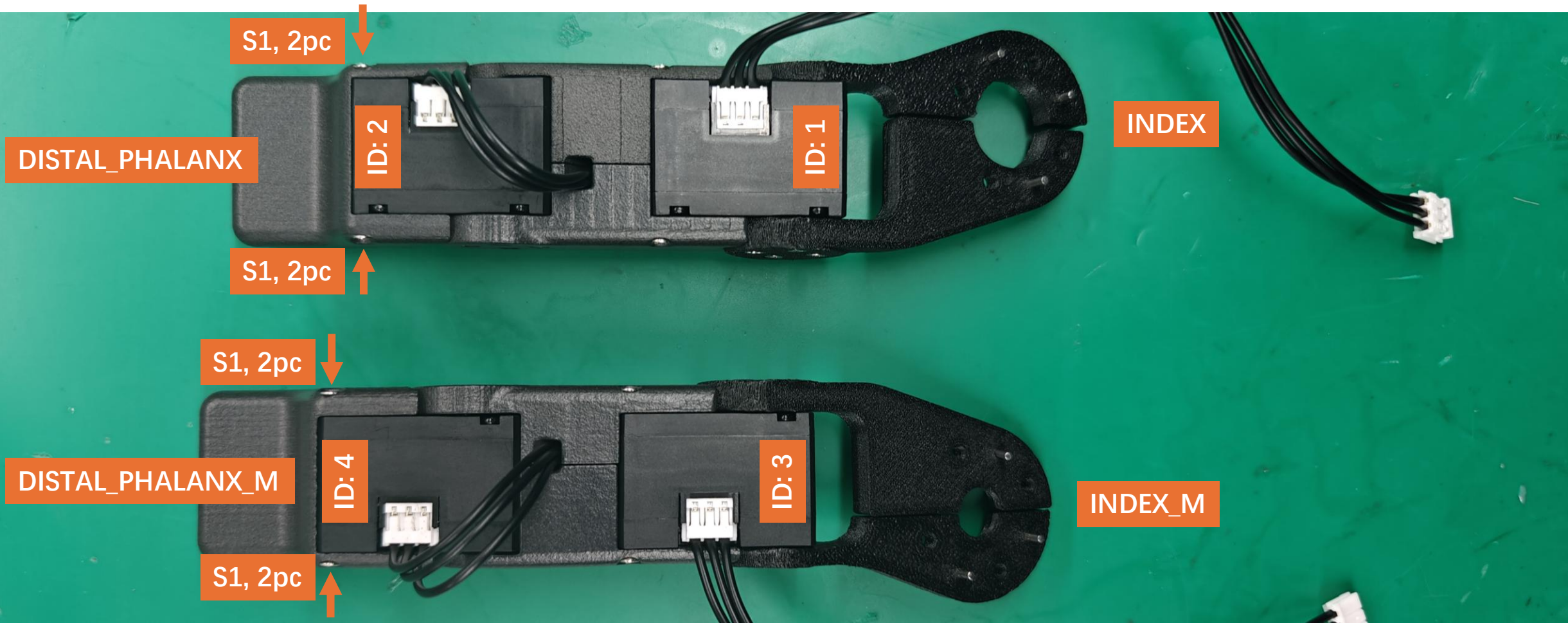




## Step 2 Index Fingers-5:

Repeat last step, assemble the **MIDDLE\_PHALANX\_M** subassembly onto the **PROXIMAL\_PHALANX\_M** subassembly to get the **INDEX\_M** subassembly.

Insert the **DISTAL\_PHALANX** and **DISTAL\_PHALANX\_M** onto **DXL ID2** and **DXL ID3** respectively as shown in the picture below, then secure with four **S1: Flat Head TRS-M2.2x12** each.



## Step 2 Index Fingers-6:

Align the 3 **P1: 1.5x10 Dowel Pin** previously installed on the **PROXIMAL\_PHALANX\_M** with the pin holes on the **SLAVE\_GEAR**, then assemble the **INDEX\_M** subassembly on to the **SLAVE\_GEAR** using four **S9: Flat Head TRS-M3x10** screws, and install one **B1: Bearing 126ZZ** as shown in the picture on the right.

### Note:

- It is usually easier to install the 126ZZ bearing in place before tightening down all four S9 screws.
- Make sure the bearing bore on the SLAVE\_GEAR is located on the bottom of the assembly.



## Step 3 Palm-1:

Assemble the **DXL ID7** onto the **PALM\_MIDHOUSING** and secure with four **S1: Flat Head TRS-M2.2x12** screws, then insert two **P2: 1.5x5 Dowel Pins** as shown in the picture below.

**IMPORTANT:** *Avoid moving the output shaft* of DXL out of its zero position during assembly.





## Step 3 Palm-2:

Assemble the **MASTER\_GEAR** onto the **DXL ID7** and secure with four **S2: Flat Head TRS-M2x8** screws as shown in the picture below.

### IMPORTANT:

Avoid moving the output shaft of DXL out of its zero position during assembly.

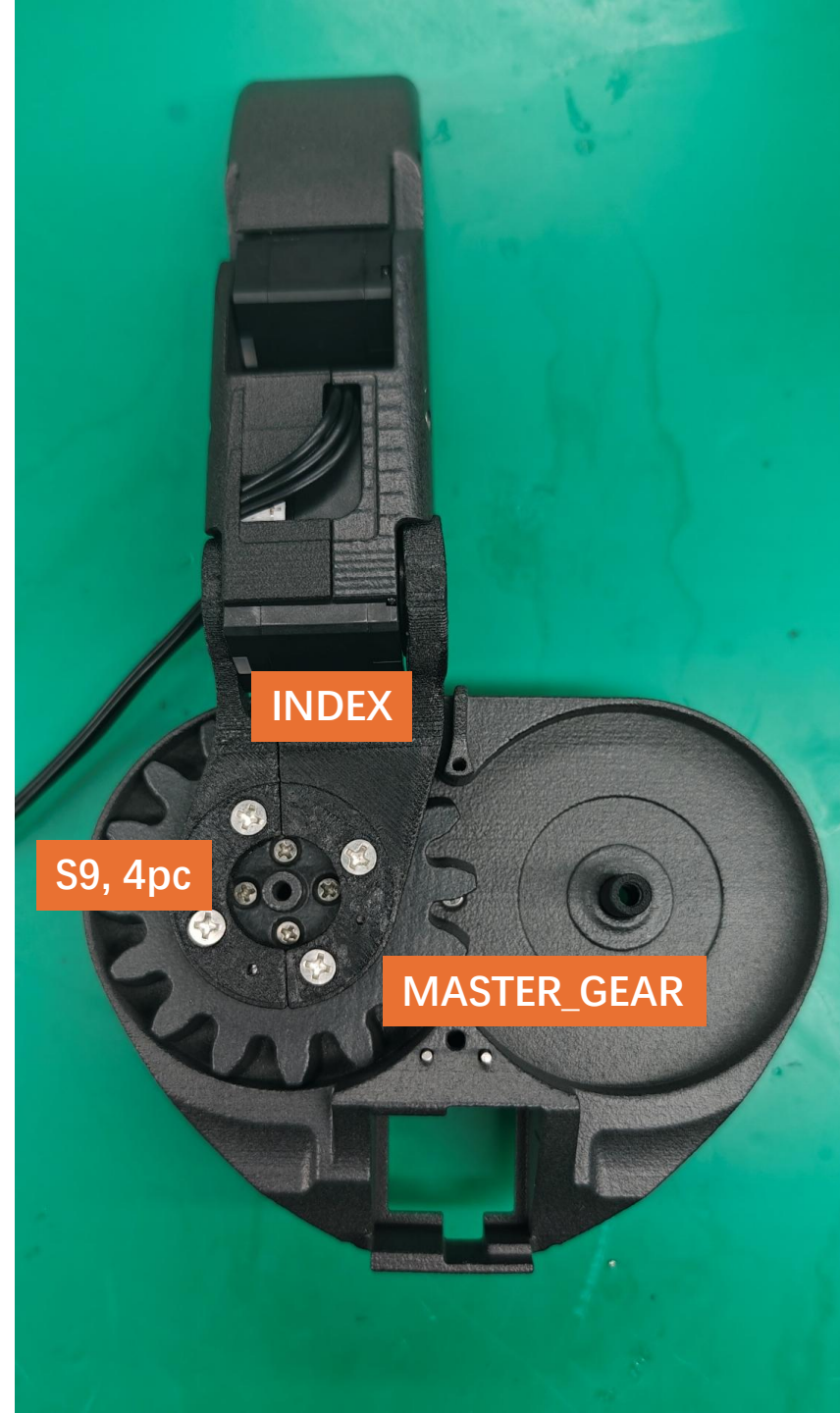


## Step 3 Palm-3:

Align the 3 **P1: 1.5x10 Dowel Pin** previously installed on the **PROXIMAL\_PHALANX** with the pin holes on the **MASTER\_GEAR**, then assemble the **INDEX** subassembly on to the **MASTER\_GEAR** using four **S9: Flat Head TRS-M3x10** screws.

### IMPORTANT:

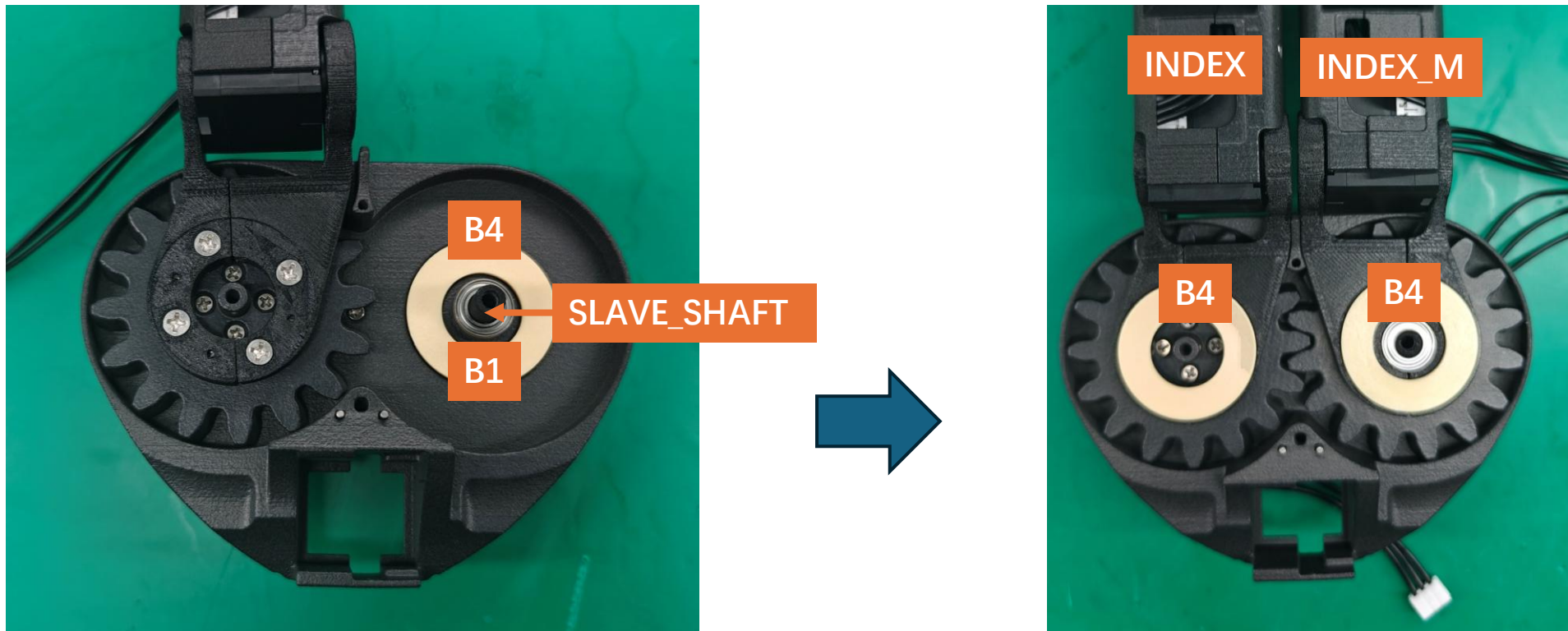
Avoid moving the output shaft of DXL out of its zero position during assembly.



## Step 3 Palm-4:

Insert the **SLAVE\_SHAFT** into the **PALM\_MIDHOUSING**, install one **B1: Bearing 126ZZ** onto the **SLAVE\_SHAFT** and place one **B4: IGUS JTM-1630-015 Thrust Bearing** in place as shown in the picture below.

Install the **INDEX\_M** subassembly on to the **SLAVE\_SHAFT**, and make sure the two index fingers are aligned as shown in the picture below. Place one **B4: IGUS JTM-1630-015 Thrust Bearing** on top of each index finger.

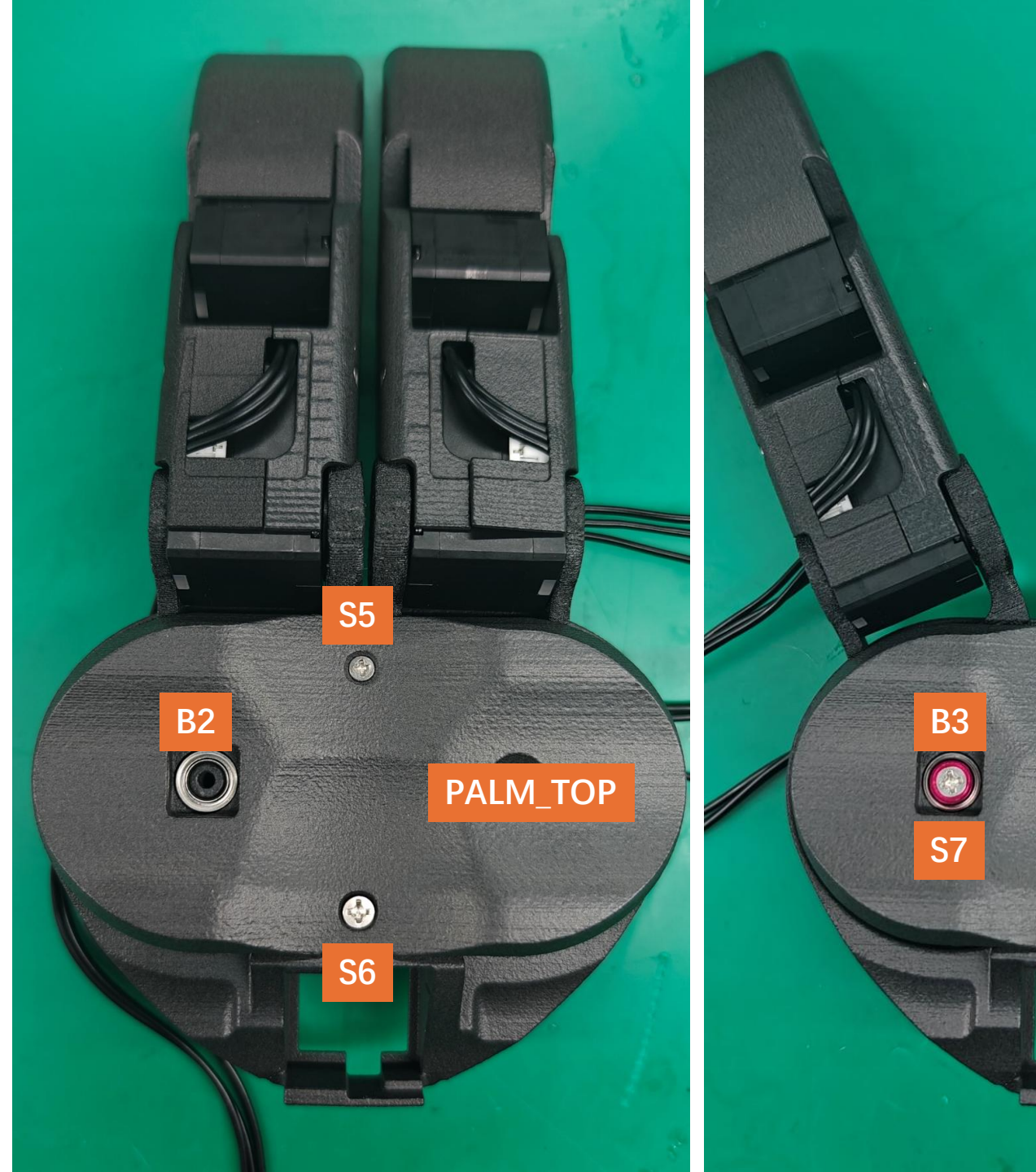




## Step 3 Palm-5:

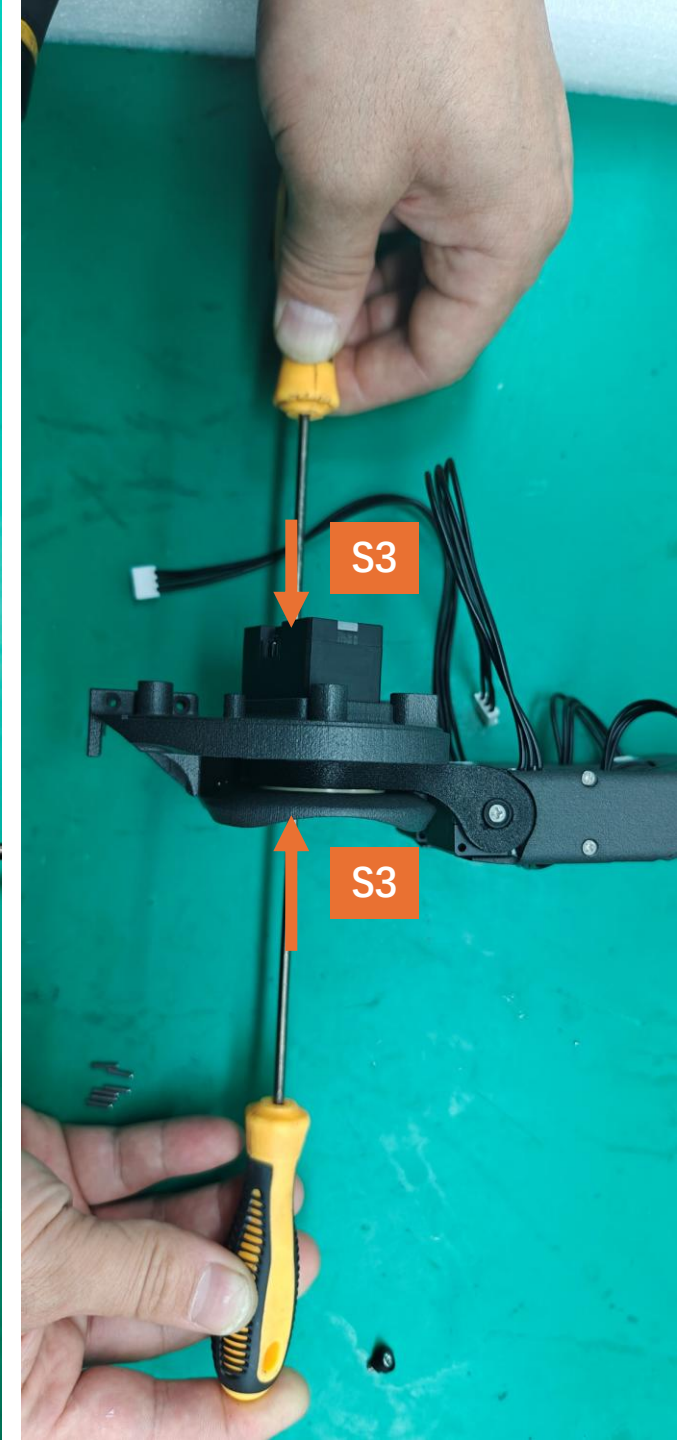
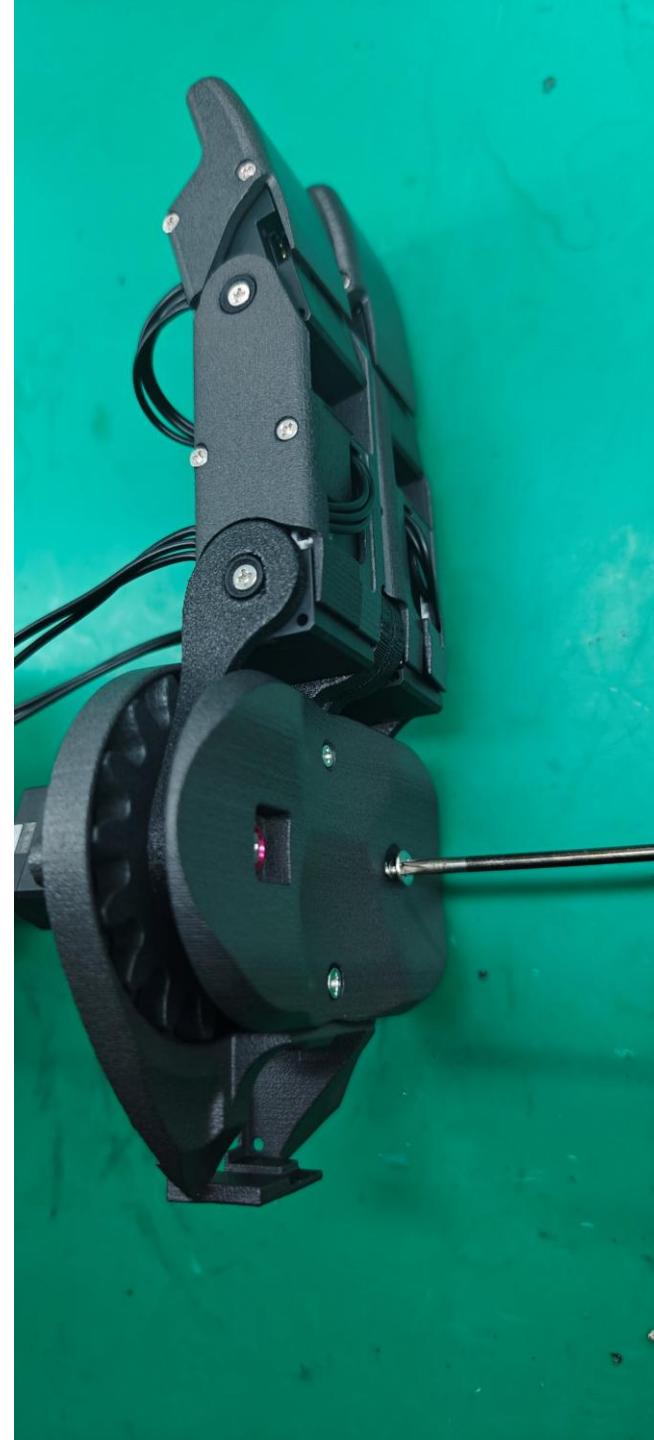
Place the **PALM\_TOP** on the **PALM\_MIDHOUSING** subassembly created in last step, secure with one **S5: Flat Head TRS-M2.5x8** screw and one **S6: Flat Head TRS-M3x12** screw.

Install the **B2: Bearing 106ZZ** bearing onto the shaft of the **MASTER\_GEAR**, then place the **B3: Metal Washer For Flat Head Screw, M3** on top of it and secure with one **S7: Flat Head TRS-M3x8** screw.



## Step 3 Palm-6:

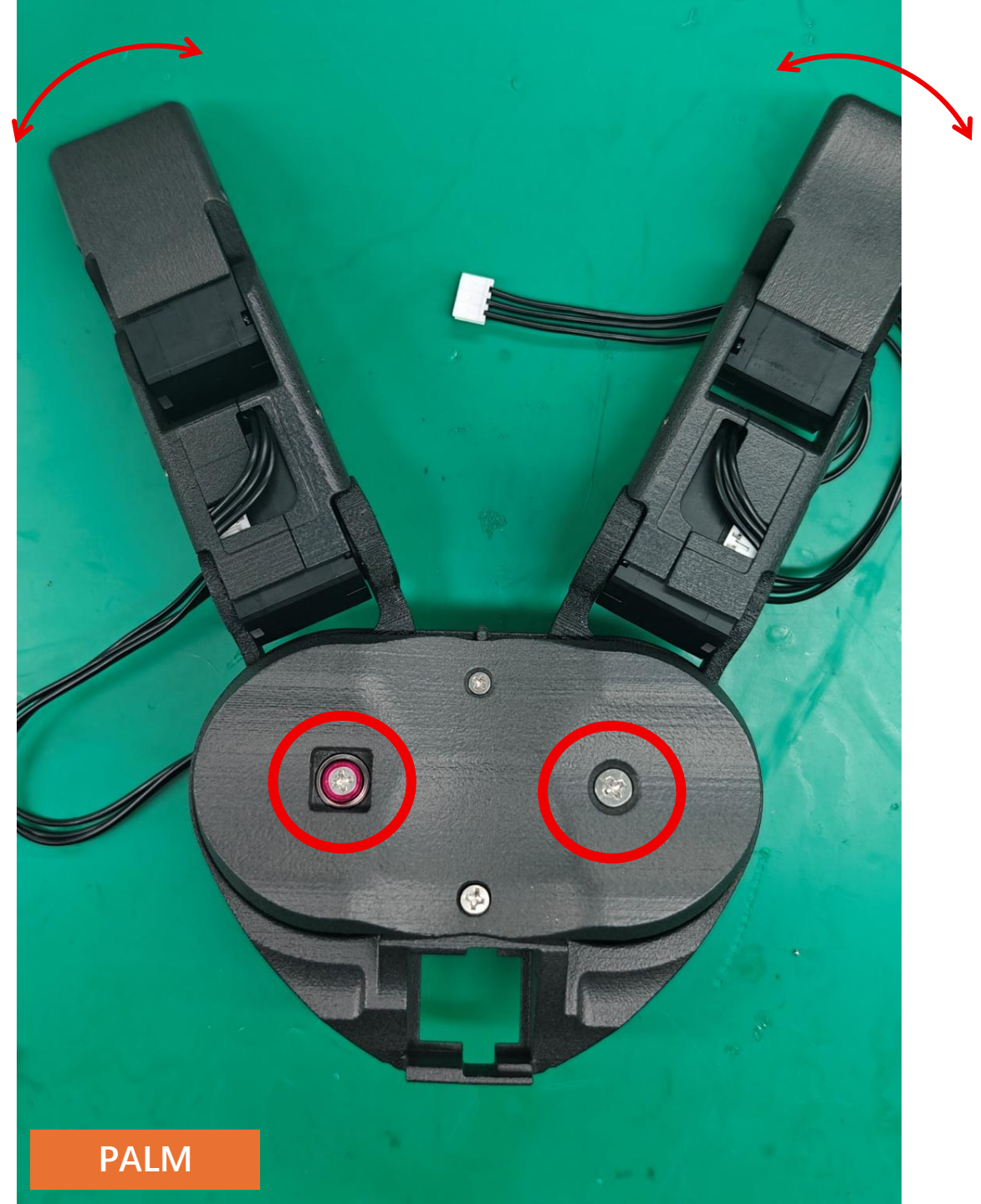
Place the **S3: Flat Head TRS-M4x12** screws onto both ends of the **SLAVE\_SHAFT**. Use two screw drivers to tighten them down from both sides.





## Step 3 Palm-7:

Make sure the two index fingers can rotate freely with relatively low friction. Slightly loosen the screws in the red circles if you find the fingers too tight.

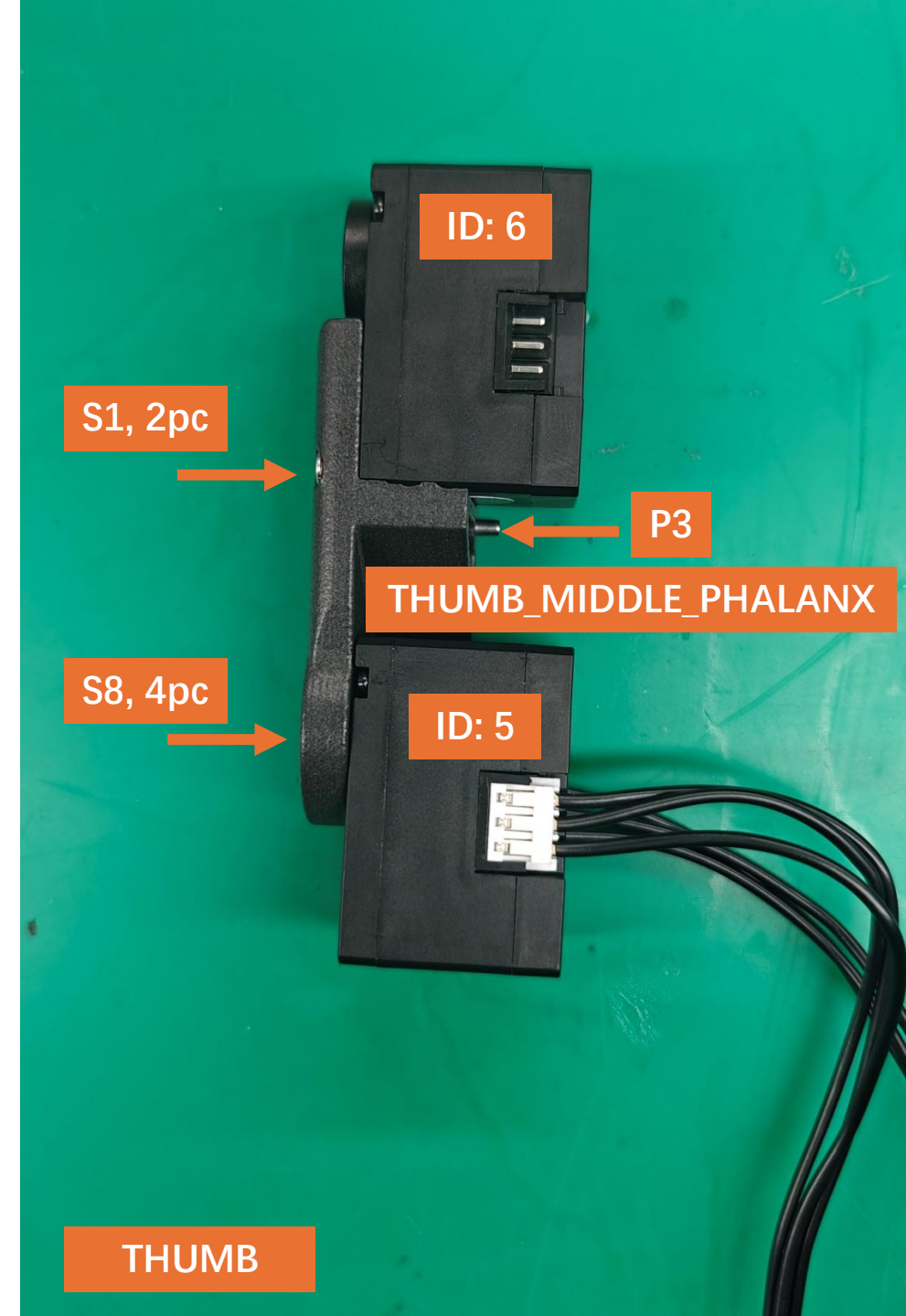


PALM



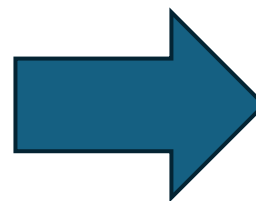
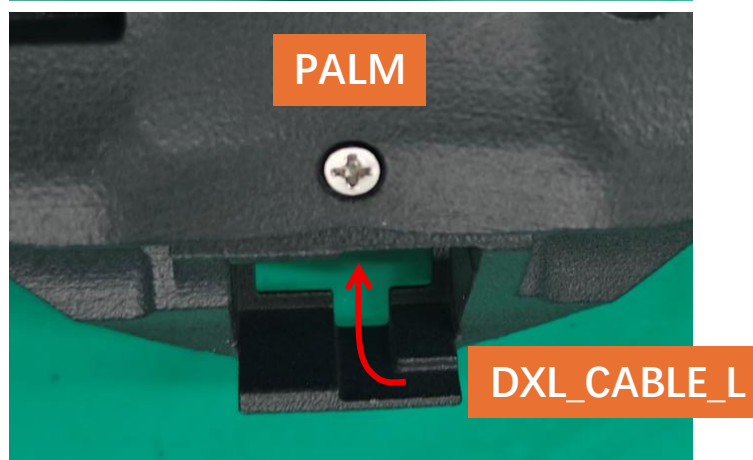
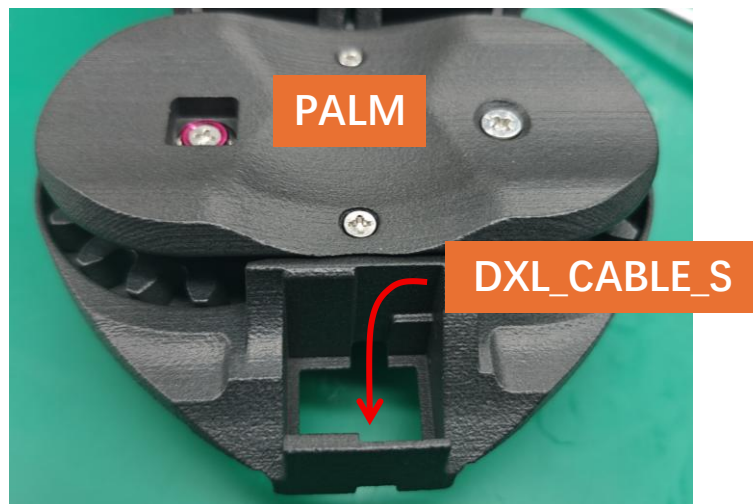
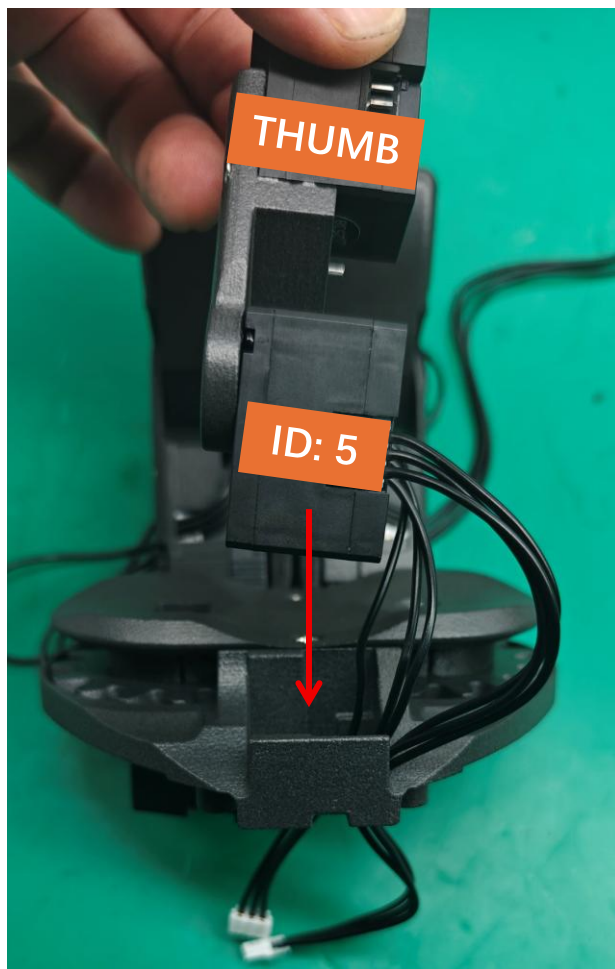
## Step 4 Thumb-1:

Assembly the **DXL ID5** and the **DXL ID6** onto half of the **THUMB\_MIDDLE\_PHALANX** with two **S1: Flat Head TRS-M2.2x12** screws and four **S8: Flat Head TRS-M2x5** screws as shown in the picture on the right. Insert one **P3: 1.5x5 Dowel Pin** into the pin hole on **THUMB\_MIDDLE\_PHALANX**.



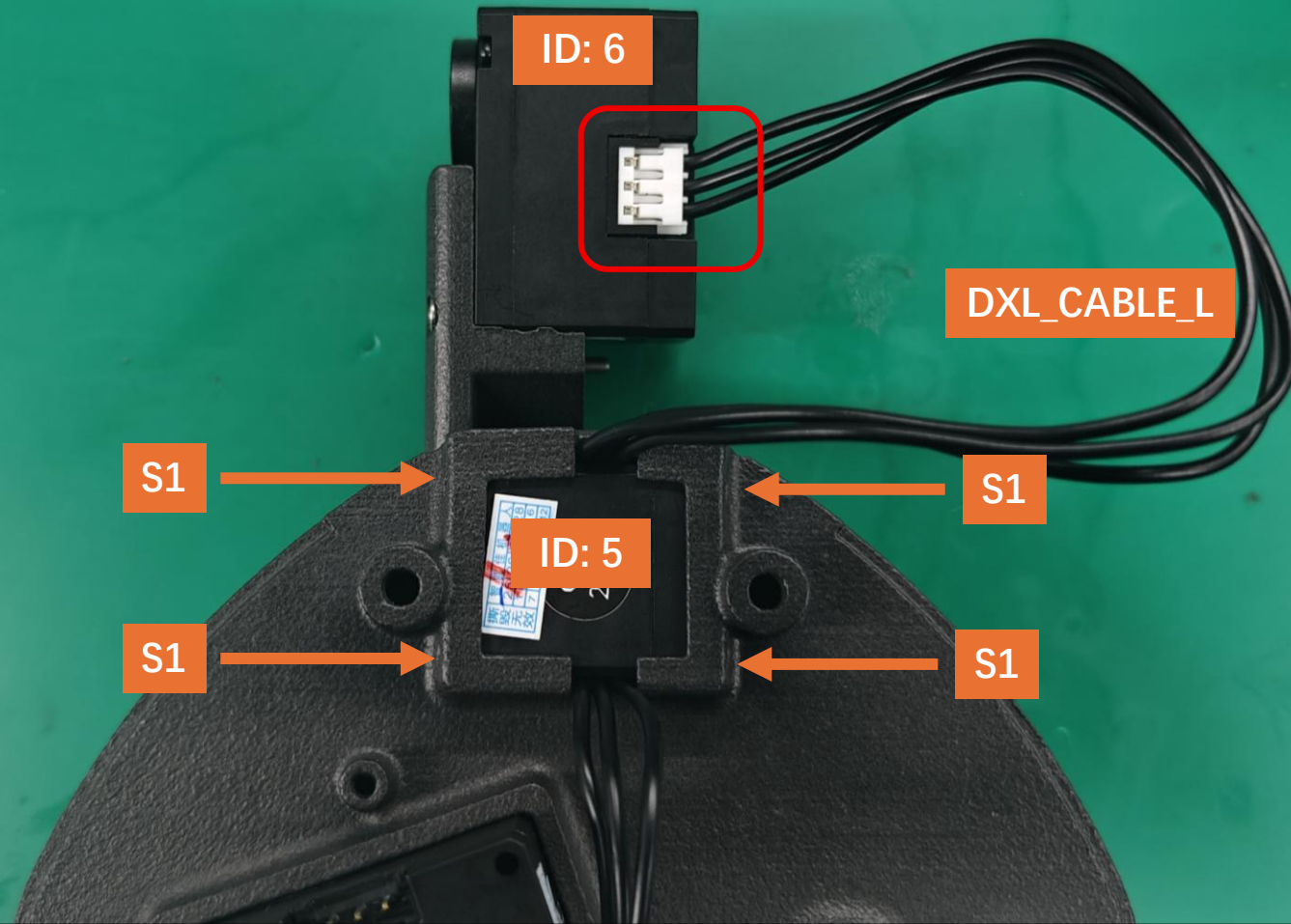
## Step 4 Thumb-2:

Insert the **DXL ID5** actuator on the **THUMB** subassembly into the housing on the **PALM** assembly. Thread the **DXL\_CABLE\_S** and **DXL\_CABLE\_L** cable connected to **DXL ID5** along the cable routing channel on the **PALM\_MIDHOUSING**. Snap the **THUMB** subassembly in place and tidy the cables are shown in the picture below.



## Step 4 Thumb-3:

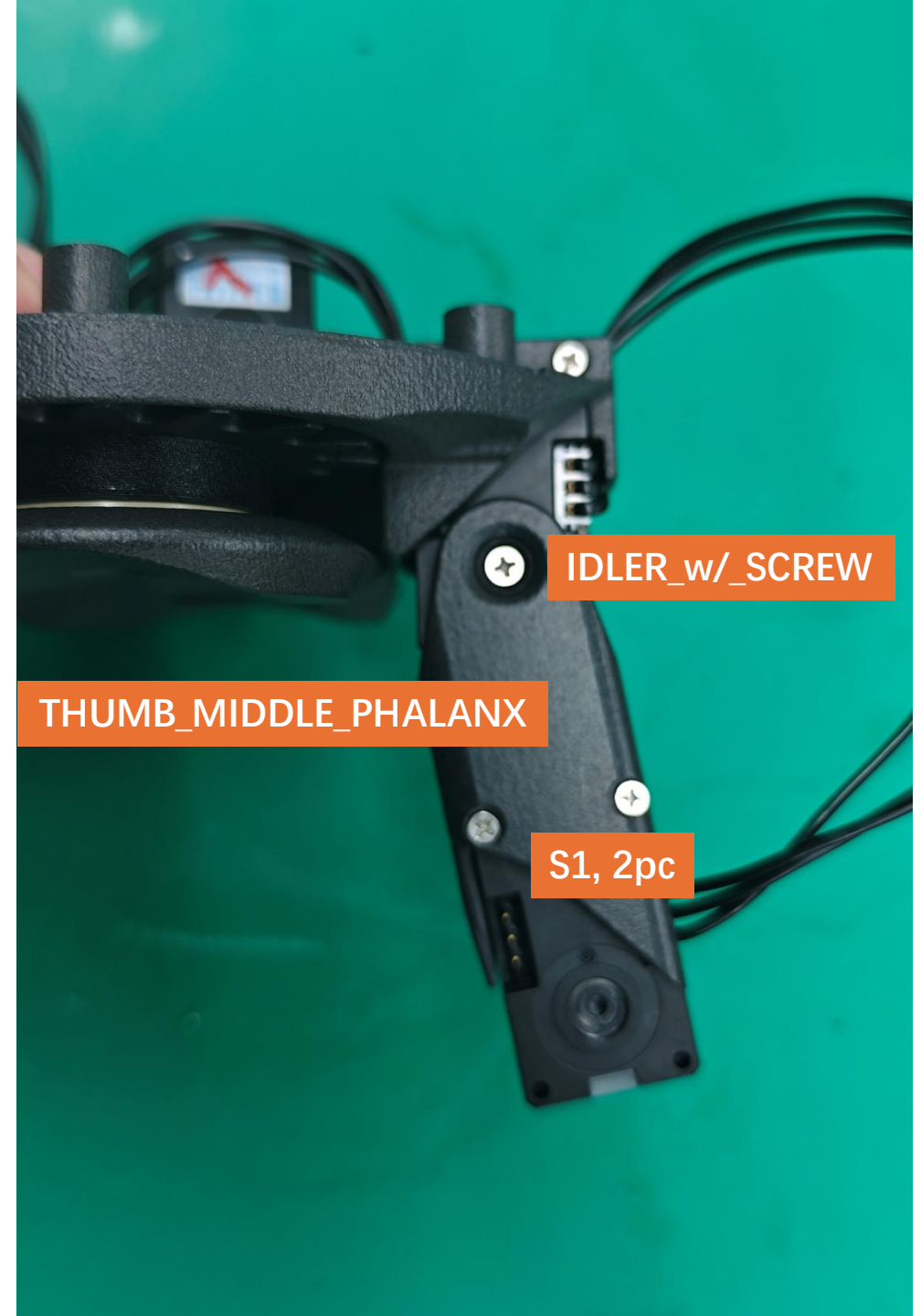
Insert the free end of the **DXL\_CABLE\_L** cable connected to **DXL ID5** into the connector on the back of **DXL ID6**. Secure the **THUMB** subassembly with four **S1: Flat Head TRS-M2.2x12** screws.





## Step 4 Thumb-4:

Assemble the other half of the **THUMB\_MIDDLE\_PHALANX** on to the **THUMB** subassembly and secure it with one **IDLER\_w/\_SCREW** and two **S1: Flat Head TRS-M2.2x12** screws.



## Step 4 Thumb-5:

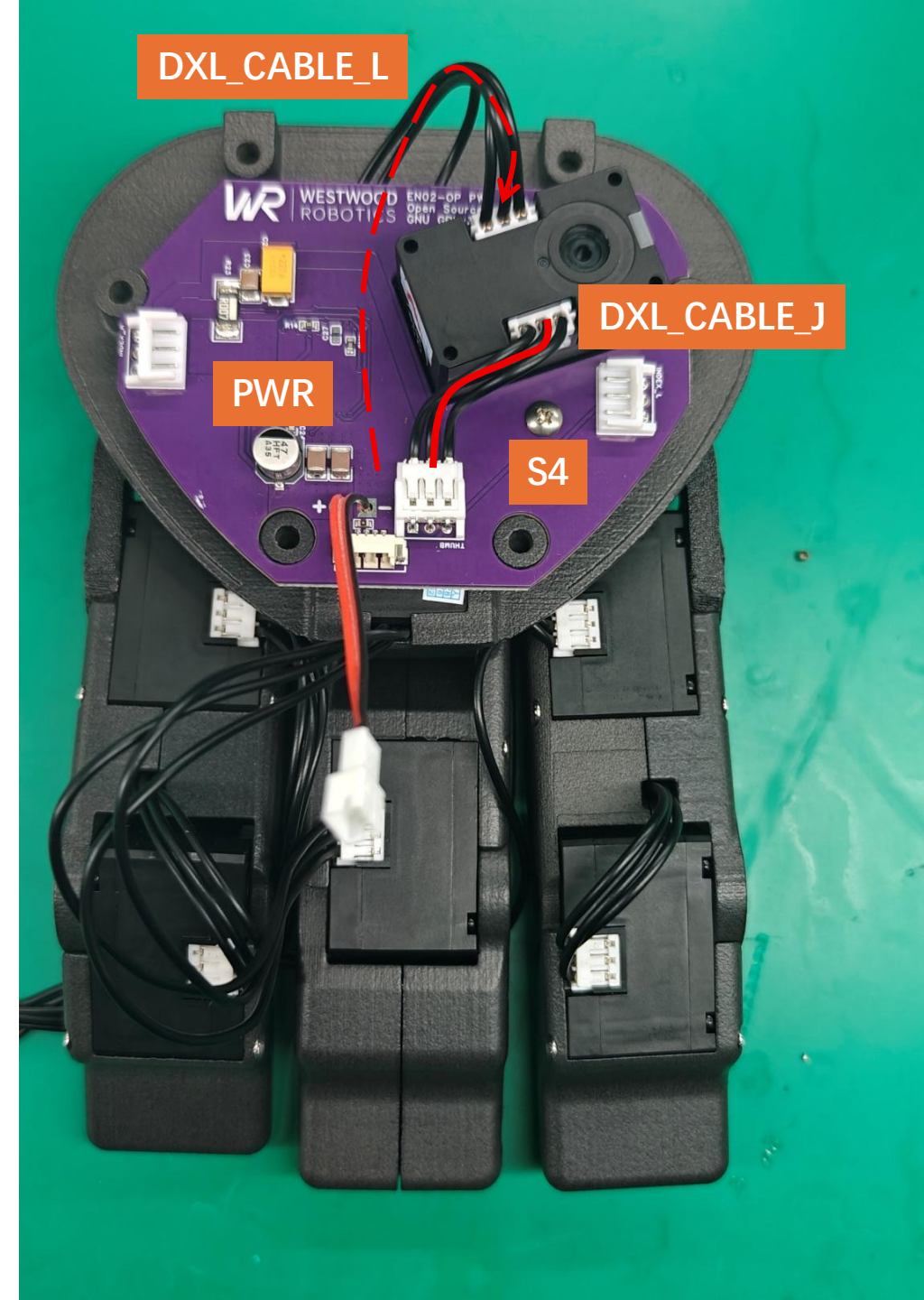
Insert two **P3: 1.5x5 Dowel Pins** into the pin holes on half of the **THUMB\_DISTAL\_PHALANX**, then assemble it onto the **DXL ID6** on the **THUMB** subassembly together with the other half of the **THUMB\_DISTAL\_PHALANX** with four **S8: Flat Head TRS-M2x5** screws and two **S11: Flat Head TRS-M2.5x24** screws.



## Step 5 Base-1:

Route the **DXL\_CABLE\_L** connected to **DXL ID5** as indicated with the red dashed line in the picture on the right, then place the **PWR** board in place and connect the **DXL ID7** in the **PALM** subassembly using the **DXL\_CABLE\_J** to the **PWR** board, as indicated with the red solid line.

Bolt down the **PWR** board with one **S4: Rounded/Button Head TRS-M2.5x6 Screw**.





## Step 5 Base-2:

Install your customized **PALM\_BASE** on to the mounting interface on your system before moving on to the next steps. We use a KBMB actuator as an example in the picture below to illustrate the concept.



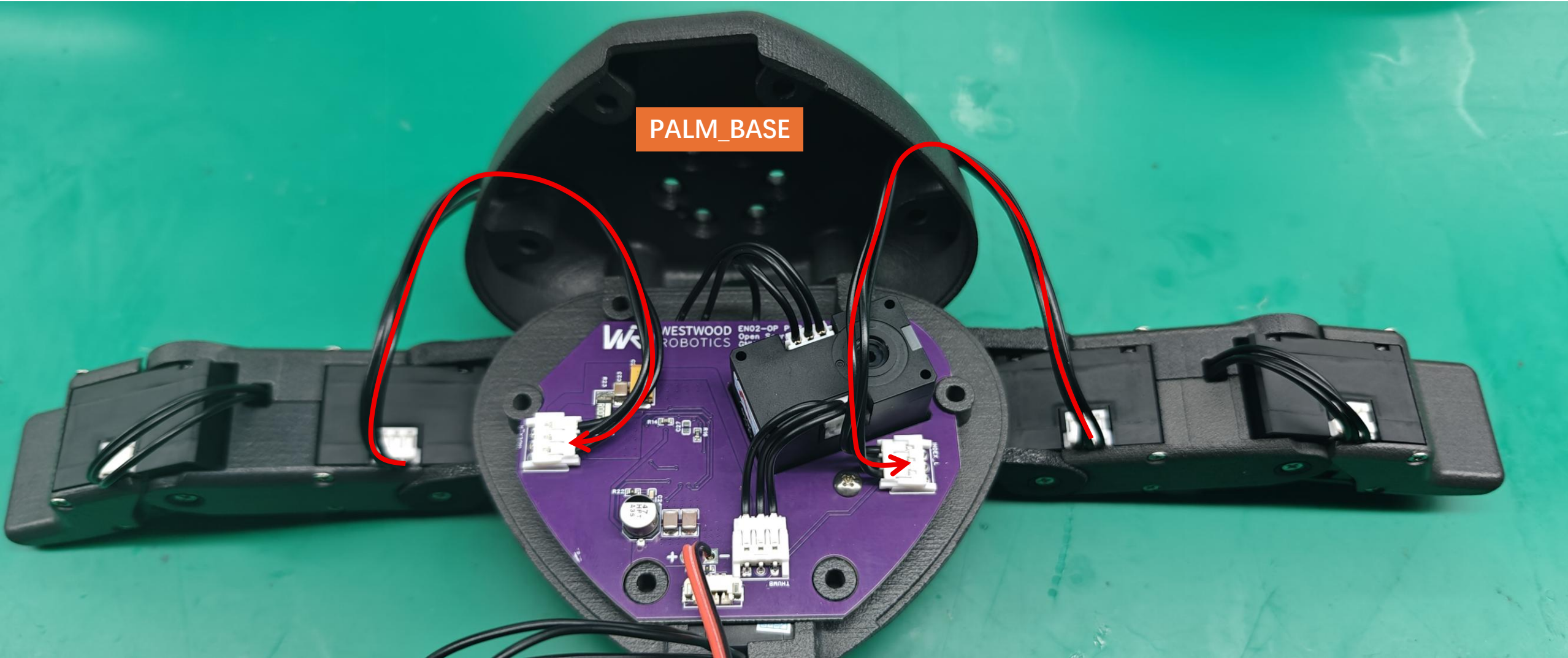
PALM\_BASE

Customized Mounting Interface

Koala BEAR Muscle Build (example)

## Step 5 Base-3:

After installing your **PALM\_BASE** on to your system, thread the **DXL\_CABLE\_L** cables connected to the two index fingers through the opening on the **PALM\_BASE**, then connect them to the ports on the **PWR** board.



## Step 5 Base-4:

Fit the **PALM** subassembly onto the **PALM\_BASE**, then secure the assembly with six **S10: Rounded/Socket Head TRS-M4x10** screws.

### IMPORTANT:

Make sure there is no interference and no signal or power cables are being pinched before installing the screws.

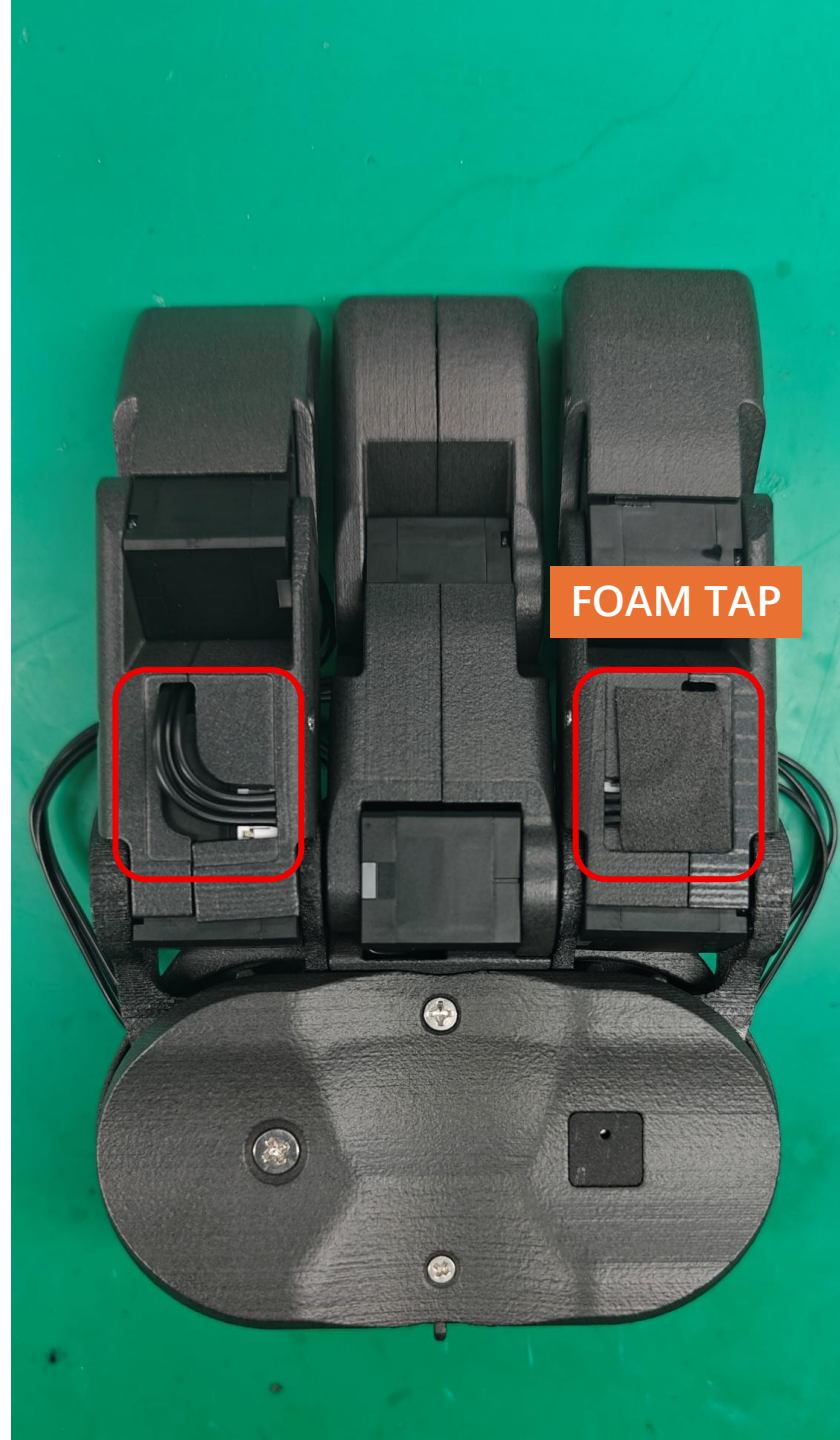




## Step 6 Protection:

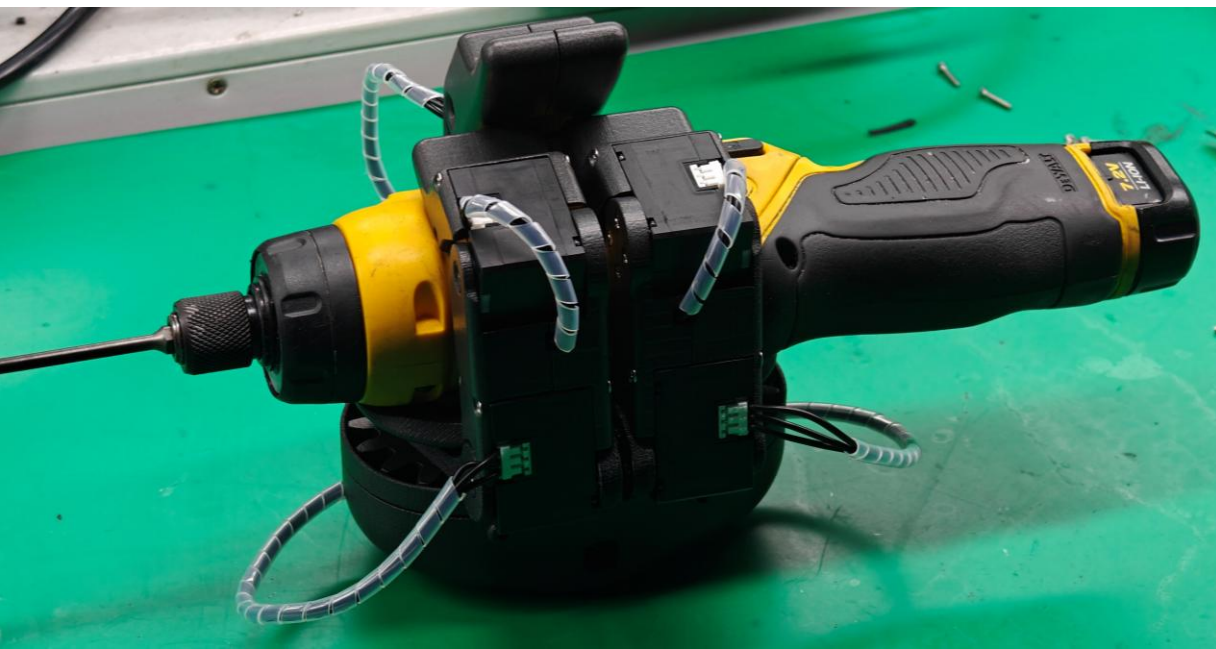
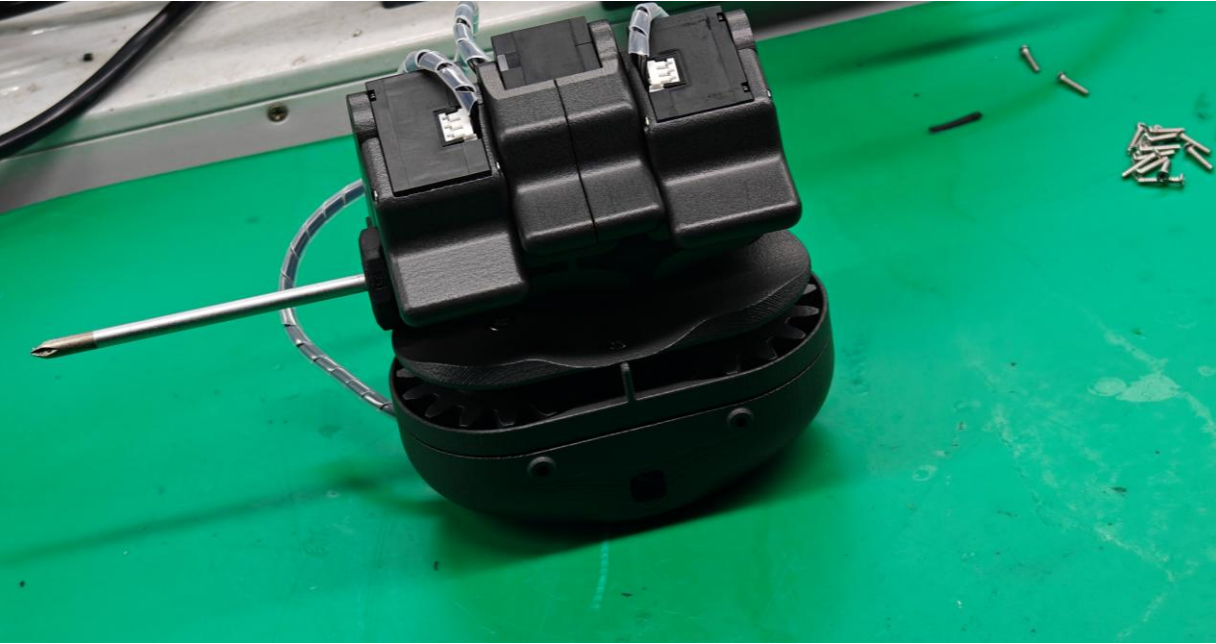
Tidy the cables in the **PROXIMAL\_PHALANX** on the two index fingers so that they lie flat along the cable channel, then cover them with **FOAM TAP** of 0.5mm or 1mm thickness.

Cover the cables with 4mm **SPIRAL WRAP** for extra tidiness and protection.





## EN02-OP Gesture and Application Examples





**THANK YOU & ENJOY!**